INCH-POUND

MIL-P-51432A(EA)
31 August 1988
SUPERSEDING
MIL-P-51432(EA)
22 February 1977

MILITARY SPECIFICATION

PLASTIC SHEET, POLYETHYLENE COPOLYMER

This specification is approved for use by U.S. Army Chemical Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of high density plastic sheet made of polyethylene copolymer.

1.2 Classification. Plastic sheets shall be of the following types as specified (see 6.1):

Type I - Polyethylene butene-1 copolymer, high density
Type II - Polyethylene hexene-1 copolymer, high density

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.1).

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-SPT-S, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. :

AMSC N/A FSC 9330

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SPECIFICATIONS

MILITARY

MIL-P-51431 - Plastic Molding and Extrusion Material, Polyethylene Copolymer, High Density

STANDARDS

FEDERAL

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies)

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issue of the documents cited in the solicitation.

ASTM STANDARDS

D 638 - Tensile Properties of Plastics
D 1238 - Flow Rates of Thermoplastics by Extrusion Plastometer
D 1505 - Density of Plastics by the Density-Gradient Technique
D 1525 - Vicat Softening Temperature of Plastics
D 1693 - Environmental Stress-Cracking of Ethylene Plastics
D 2240 - Rubber Property-Durometer Hardness

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103).

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Material. The plastic sheet shall be made from high density polyethylene copolymer conforming to type I or type II of MIL-P-51431 as specified (see 6.1). No additives, coloring material, regrind material, processing lubricants, alcohols, glycols, sugars, organic acids, other compounds containing hydroxyl groups, mercaptans, amines, or other foreign matter shall be added to the virgin polyethylene copolymer or used during the manufacture of the plastic sheet. The contractor shall certify that this requirement has been met.

3.2 Infrared spectrum characteristics. The plastic sheet shall have the infrared spectrum characteristics specified in 4.4.4.1 when tested as specified therein.

3.3 Water content. The plastic sheet shall contain no more than 0.01 percent by weight water when tested as specified in 4.4.4.2.

3.4 Dimensions. The length, width, and thickness of the plastic sheet shall be as specified in the contract or order (see 6.1).

3.5 Physical characteristics. The plastic sheet shall conform to the physical characteristics of table I when tested as specified in 4.4.4.3.

TABLE I. Physical characteristics

Characteristic	Requirement			
	Type I		Type II	
	Minimum	Maximum	Minimum	Maximum
Density, grams per cubic centimeter	0.935	0.945	0.940	0.950
Environmental stress-cracking resistance, hours, F_{50}, full strength Igepal CO-630 as reagent	1000*	----	1000*	----
Flow rate, grams per 10 minutes	----	2.0	----	2.0
Hardness, type D durometer	60	66	64	70
Tensile strength at yield, psi, 2 inches per minute	2400	----	2400	----
Elongation at break, percent 2 inches per minute	300	----	300	----
Vicat softening point, °F	257	----	245	----

*Environmental stress-cracking resistance test shall not be performed on sheets having a thickness greater than 0.07 inch.

3.6 Workmanship. The plastic sheet shall have a smooth finish and shall be free of cracks, blisters, bubbles, discoloration, craze, surface scratches that form definite indentations, and other defects that could affect appearance or serviceability.

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3.7 Preproduction. Prior to the start of regular production, a preproduction sample of plastic sheet shall be produced in accordance with this specification for examination and test (see 4.3).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

4.3 Preproduction inspection.

4.3.1 Lotting. A preproduction lot of 1,000 square feet of plastic sheet of each thickness specified (see 6.1) shall be manufactured using the same methods, materials, equipment, and processes as will be used during regular production.

4.3.2 Sampling.

4.3.2.1 For examination. Twenty-five sheets shall be randomly selected from the preproduction lot.

4.3.2.2 For test. One sample of plastic sheet no less than 6 inches long and as wide as the full width of the extrusion shall be cut from the extrusion line at the beginning, the middle, and the end of the preproduction lot.

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4.3.3 Inspection procedure.

4.3.3.1 For examination. The sample sheets selected in 4.3.2.1 shall be examined for the defects listed in 4.4.3.2.

4.3.3.2 For test. The sample shall be tested as specified in 4.4.3.3.

4.3.4 Acceptance/rejection criteria. The preproduction samples shall pass the examinations and tests specified in 4.3.3 to be acceptable. The acceptance number shall be zero. The contractor shall obtain written approval from the contracting officer prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the plastic sheet of one type produced by one manufacturer, at one plant, on one extruder, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous.

4.4.2 Sampling.

4.4.2.1 For examination of packaging. Sampling for examination of packaging shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For examination of plastic sheet. Sampling for examination of plastic sheet shall be conducted in accordance with MIL-STD-105.

4.4.2.3 For physical characteristics tests. One sample of plastic sheet no less than 6 inches long and as wide as the full width of the extrusion shall be cut from the extrusion line at start up, after every 250 linear feet of production, after a line shutdown and start up, on changing lots of polyethylene resin, and at the end of the production. The samples shall be marked to identify the lot represented, the date and hour that they were taken, and the extrusion line that they were taken from if more than one line is involved.

4.4.2.4 For infrared spectrum characteristics and water content tests. If more than one lot of sheet is produced from a single lot of resin, one sample of plastic sheet no less than 6 inches long and as wide as the full width of the extrusion shall be cut from the first sheet lot and one from the last sheet lot produced from the resin lot. If more than one lot of resin is used to produce a single lot of sheet, one sample of plastic sheet no less than 6 inches long and as wide as the full width of the extrusion shall be taken from each resin lot represented. If one lot of resin is used to produce one lot of sheet, one sample shall be taken at the beginning and one sample at the end of the lot.

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4.4.3 Inspection procedure.

4.4.3.1 For examination of packaging. The sample unit shall be one filled shipping container. Sample shipping containers and the packaging thereof shall be examined for the following defects using an AQL of 2.5 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged
(e) Marking incorrect, missing, or illegible

4.4.3.2 For examination of plastic sheet. The sample unit shall be one sheet having dimensions specified in the contract or order (see 6.1). Sample sheets shall be examined for the following defects using an AQL of 1.0 defects per hundred units:

(a) Length of sheet not as specified
(b) Width of sheet not as specified
(c) Thickness of sheet not as specified
(d) Workmanship (see 3.6)

4.4.3.3 For test. Each sample taken in 4.4.2.3 and 4.4.2.4 shall be tested as specified in 4.4.4 except that only one sample from the first and one sample from the last lot of sheets produced from each lot of plastic resin shall be tested as specified in 4.4.4.1. Two specimens from each sample shall be separately tested for environmental stress-cracking resistance, tensile strength at yield, and elongation at break. One of the specimens shall be die cut from the sample in the direction of extrusion. The other specimen shall be die cut from the sample at an angle of 90° to the direction of extrusion. Failure of either specimen in any of the above tests shall be cause for rejection of the lot represented. Failure of any other test by the sample shall be cause for rejection of the lot represented.

4.4.4 Tests. Tests shall be conducted as follows:

4.4.4.1 Infrared spectrum characteristics. Determine infrared spectrum characteristics in accordance with the test procedure for infrared spectrum characteristics in MIL-P-51431.

4.4.4.2 Water content. Using a solids moisture analyzer (see 6.2) and a cylinder of dry nitrogen, regulate the nitrogen pressure to the dryer and the instrument at 5 pounds per square inch (psi). Before installing the electrolytic cell, purge the dryer and the instrument for 10 to 15 minutes. Turn off the gas and install the cell, making sure that the cell posts are properly oriented with nozzle fittings inside the cell case. Set the timer in the "Off" position and plug in the power cord. The "Proceeding" lamp will light for approximately 2 minutes and then will shut off automatically. At this time, the "Completed" lamp will light and remain on. With the sample oven sealed and

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the nitrogen pressure at 5 psi, allow the instrument to dry down for 1/2 hour. Set and adjust the oven temperature to 130°C. Before running a sample, check the electrical zero of the integrator by pressing the "Zero" push button and observing the integrator count rate. The "Zero Adjust" control should be set so that the count rate is about 2 to 3 micrograms per minute. Insert the sample boat to be used in the instrument oven and allow it to dry for 30 minutes. Remove the empty boat from the oven for 10 to 15 seconds and then replace it in the oven and record the count for 30 minutes. Zero the integrator. Working as quickly as possible to avoid gaining or losing moisture, transfer approximately 0.5 gram of sample from a previously weighed weighing bottle to the empty sample boat which has been kept dry, remove the oven cap, insert the sample boat into the heated area using tweezers, replace the oven cap and turn to lock, and then turn the timer on. Set the timer for the time limit desired. For polyethylene, 30 minutes is usually sufficient time to drive off all the water. For the first sample, set the timer for maximum time in order to establish the proper running time needed to drive out all the water. Allow the instrument to run until the integrator runs very slowly. Then turn the timer off and record the count. Once the proper running time is established, the timer can be set to automatically turn the oven and the integrator off and to turn the oven blower on. The blower will continue to run until the oven temperature has been lowered to ambient. A new sample should not be placed in the instrument until the "Completed" lamp lights. Calculate the percent by weight water as follows:

$$\text{Percent water} = \frac{(A - B)(10^{-4})}{W}$$

where: A = Microgram count for sample,
B = Microgram count for empty sample boat, and
C = Weight of sample in grams.

4.4.4.3 Physical characteristics. Determine the physical characteristics of the sample specified in table I in accordance with the test methods specified in table II.

TABLE II. Test methods for physical characteristics

Characteristic	ASTM test method
Density	D 1505
Environmental stress-cracking resistance	D 1693, condition B
Flow rate	D 1238, procedure A, condition F
Hardness, type D durometer	D 2240
Tensile strength at yield	D 638, type IV specimen, speed C
Elongation at break	D 638, type IV specimen, speed C
Vicat softening point	D 1525

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5. PACKAGING

5.1 Interplant shipment (see 6.3). A quantity of plastic sheets as specified in the contract or order (see 6.1) shall be unit packed and packed in a manner to provide adequate protection from physical damage or deformation and exposure to light from the supply source to the first receiving activity for immediate use or further processing. Sheets shall be packed in a lay flat configuration with full support to protect against flexing or bending during shipment and storage. Shipping containers shall be in compliance with the rules and regulations applicable to the mode of transportation.

5.2 Marking. Shipping containers shall be marked in accordance with Fed. Std. No. 123 and shall include the contractor's name, address, contract number, and date of manufacture.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification
(b) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2)
(c) Type of material required (see 1.2 and 3.1)
(d) Length, width, and thickness required (see 3.4, 4.3.1, and 4.4.3.2)
(e) Quantity of sheets per shipping container (see 5.1)
(f) Preproduction:

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for Government to notify the contractor whether or not to proceed with production.

6.2 Solids moisture analyzer. A DuPont type 26-321A moisture analyzer has been found satisfactory for use in the water content test.

6.3 Interplant shipment. Unit packing and packing for interplant shipment is designated for supplies and materials that do not enter the military supply system. Typical interplant shipments are those from a vendor to a subcontractor or a prime contractor, or from a subcontractor to a prime contractor, or from a vendor to a contractor to a military arsenal or plant.

6.4 Extrusion of plastic sheet. One combination of equipment and operating conditions that may be used for extruding plastic sheet is described in the appendix to this specification.

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6.5 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E 29.

6.6 Subject term (key word) listing.

Polyethylene butene-1 copolymer
Polyethylene hexene-1 copolymer

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 9330-A012

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APPENDIX

EXTRUSION OF PLASTIC SHEET

10. SCOPE

10.1 This appendix describes equipment and operating conditions that have been used in the extrusion of plastic sheet. This information is furnished with the distinct understandings:

(a) That the contractor is not required to follow the method.

(b) That the requirements of this specification must be met regardless of the method of manufacture.

(c) That the Government makes no warranty that plastic sheet made with equipment described and under the conditions described will meet the requirements of the specification.

20. EQUIPMENT

20.1 Extruder. Use an extruder that is 3-1/2 inches in diameter and has a 24 to 1 L/D screw and a 75-horsepower drive motor.

20.2 Screw. Use a screw with nine turns feed 0.450 inch deep, nine turns transition - constant taper, six turns meter 0.180 inch deep, and a pitch of 3-1/2 inches.

20.3 Die. Use a 40-inch wide modified inverted coat hanger with choke bar as manufactured by the National Rubber Machinery Company, Akron, Ohio.

20.4 Sheet train. The sheet train should consist of a five-roll (12-inch diameter) polishing stand, a 21-foot cooling conveyor, rubber tension rolls, and a power shear.

30. OPERATING CONDITIONS

30.1 The extruder temperature profile should be as listed in table III.

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TABLE III. Extruder temperature profile

Characteristic	Condition
Zone 1 temperature	515°F
Zone 2 temperature	510°F
Zone 3 temperature	475°F
Zone 4 temperature	525°F
Zone 5 temperature	460°F
Die temperature	430°F
Stock temperature	480°F
Roll temperature, top	190°F
Roll temperature, middle	195°F
Roll temperature, bottom	210°F
Roll temperature, No. 4	185°F
Roll temperature, No. 5	180°F
Motor load, amps	110
Line speed, inches per minute	6.25
Screw rpm	30
Head pressure, psig	3200
Extruder output, pounds per hour	127
Die opening, inches	0.320

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-51432A(EA)	PLASTIC SHEET, POLYETHYLENE COPOLYMER

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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